

YUNYONG KO

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Research Interests	My research interests mainly lie in data mining and machine learning on a various types of data (e.g., graph, hypergraph, text, image, and time-series), with a special focus on exploring knowledge for real-world applications.	
Educational Background	Hanyang University, Seoul, Korea • Ph.D. in Computer Science - Thesis: Effective Approaches to Distributed Deep Learning: Methods, Analyses, and Evaluation - Advisor: Prof. Sang-Wook Kim Hanyang University, Seoul, Korea • B.S. in Computer Science	Sep. 2013 – Aug. 2021 Mar. 2009 – Aug. 2013
Work Experiences	Chung-Ang University, Seoul, Korea • Assistant Professor, School of Computer Science and Engineering University of Illinois at Urbana-Champaign, Urbana, IL, USA • Postdoctoral Researcher, Department of Computer Science - Topic: Large-Scale Hypergraph Learning for Real-World Applications - Advisor: Prof. Hanghang Tong Hanyang University, Seoul, Korea • Postdoctoral Researcher, Department of Computer Science - Topic: Optimization Technique for Large-Batch DNN Training - Advisor: Prof. Sang-Wook Kim The Pennsylvania State University, University Park, PA, USA • Visiting Scholar, College of Information Sciences and Technology - Topic: Data Parallelism Approach for Distributed Deep Learning - Advisor: Prof. Dongwon Lee	Mar. 2024 – Present May. 2022 – Feb. 2024 Sep. 2021 – April. 2022 Oct. 2019 – Feb. 2020
Awards & Honors	Received the Scholarship and Teaching for Engineering Postdocs (STEP) • Grainger College of Engineering (GCOE), University of Illinois at Urbana-Champaign Selected as One of the Best-Ranked Papers • IEEE International Conference on Data Mining (IEEE ICDM) Selected as One of the Spotlight Presentations • ACM International Conference on Information and Knowledge Management (ACM CIKM) Received the Outstanding Ph.D. Dissertation Award • Research Institute of Industrial Science, Hanyang University Received the ACM SIGAPP Student Travel Award • ACM Symposium on Applied Computing (ACM SAC) Awarded the NAVER Ph.D. Fellowship • Naver Corporation Domestic Conference/Journal Awards • Best Paper Awards: KIPS Spring Confernece (2021, 2023), Journal of KIISE (2017) • Best Presentation Award: KIISE KCC (2017)	2023 2021 2021 2021 2019 2017

Preprinted Papers

- [3] Revisiting Evaluation of Knowledge Graph Completion: Predictive Sharpness and Popularity-bias Robustness
SooHo Moon and Yunyong Ko
Under Review at One of Top-Tier DM Conferences
- [2] Learning High-Order Dynamics of Real-World Networks
Daeun Lee, Song Kyung You, Yunyong Ko, and Sang-Woo Kim
Under Review at One of Top-Tier DM Conferences
- [1] AGNES: Accelerating Storage-based GNN Training for Web-scale Graphs
Myung-Hwan Jang, Jeong-Min Park, Yunyong Ko, and Sang-Wook Kim
Under Review at One of Top-Tier DM Conferences

International Conference and Journal Papers (* indicates equal contributions)

- [16] CROWN: A Novel Approach to Comprehending Users' Preferences for Accurate Personalized News Recommendation
Yunyong Ko, Seongeun Ryu, and Sang-Wook Kim
WWW'25 – *ACM Web Conference*
Selected as an Oral Presentation of WWW 2025
- [15] HyGEN: Regularizing Negative Hyperedge Generation for Accurate Hyperedge Prediction
Song Kyung Yu, Da Eun Lee, Yunyong Ko, and Sang-Wook Kim
WWW'25-Short – *ACM Web Conference*
- [14] Enhancing Hyperedge Prediction with Context-Aware Self-Supervised Learning
Yunyong Ko, Hanghang Tong, and Sang-Wook Kim
TKDE – *IEEE Transactions on Knowledge and Data Engineering*
SCIE Journal (Q1: IF Top 1.8%)
- [13] HearHere: Mitigating Echo Chambers in News Consumption through an AI-based Web System
Youngseung Jeon, Jaehoon Kim, Sohyun Park, Yunyong Ko, Seongeun Ryu, Sang-Wook Kim, and Kyungsik Han
CSCW'24 – *ACM Conference on Computer-Supported Cooperative Work and Social Computing*
- [12] SAGE: A Storage-Based Approach for Scalable and Efficient Sparse Generalized Matrix-Matrix Multiplication
{Myung-Hwan Jang*, Yunyong Ko*}, Hyuck-Moo Gwon, Ik-Hyeon Jo, Yongjun Park, and Sang-Wook Kim
CIKM'23 – *ACM International Conference on Information and Knowledge Management*
- [11] KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction
Yunyong Ko, Seongeun Ryu, Soeun Han, Youngseung Jeon, Jaehoon Kim, Sohyun Park, Kyungsik Han, Hanghang Tong and Sang-Wook Kim
WWW'23 – *ACM Web Conference*
- [10] RealGraph^{GPU}: A High-Performance GPU-Based Graph Engine Toward Large-Scale Real-World Network Analysis
Myung-Hwan Jang, Yunyong Ko, Dongkyu Jeong, Jeong-Min Park, and Sang-Wook Kim
CIKM'22-Short – *ACM International Conference on Information and Knowledge Management*
- [9] Not All Layers Are Equal: A Layer-Wise Adaptive Approach Toward Large-Scale DNN Training
Yunyong Ko, Dongwon Lee, and Sang-Wook Kim
WWW'22 – *ACM Web Conference*
- [8] D-FEND: A Diffusion-Based Fake News Detection Framework for News Articles Related to COVID-19
Soeun Han, Yunyong Ko, Yusim Kim, Heejin Park, Seongsu Oh, and Sang-Wook Kim
SAC'22 – *ACM Symposium on Applied Computing*

- [7] **MASCOT: A Quantization Framework for Efficient Matrix Factorization in Recommender Systems**
 {Yunyong Ko*, Jae-Seo Yu*}, Hong-Kyun Bae, Yongjun Park, Dongwon Lee, and Sang-Wook Kim
ICDM'21 – IEEE International Conference on Data Mining
Selected as One of the Best-ranked Papers of ICDM 2021 for Fast-track Journal Invitation
- [6] **ALADDIN: Asymmetric Centralized Training for Distributed Deep Learning**
 Yunyong Ko, Kibong Choi, Hyunseung Je, Dongwon Lee, and Sang-Wook Kim
CIKM'21 – ACM International Conference on Information and Knowledge Management
Selected as a Spotlight Presentation of CIKM 2021
- [5] **An In-Depth Analysis of Distributed Training of Deep Neural Networks**
 Yunyong Ko, Kibong Choi, Jiwon Seo, and Sang-Wook Kim
IPDPS'21 – IEEE International Parallel and Distributed Processing Symposium
- [4] **Influence Maximization for Effective Advertisement in Social Networks: Problem, Solution, and Evaluation**
 Suk-Jin Hong, Yunyong Ko, Moon-Jeung Joe, and Sang-Wook Kim
SAC'19 – ACM Symposium on Applied Computing
- [3] **Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach**
 Yunyong Ko, Kyung-Jae Cho, and Sang-Wook Kim
Information Sciences
SCIE Journal (Q1: IF Top 5%)
- [2] **Influence Maximization in Social Networks: A Target-Oriented Estimation**
 Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
Journal of Information Science
SCIE Journal (Q2)
- [1] **Accurate Path-Based Influence Maximization in Social Networks**
 Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
WWW'16-Short – ACM Web Conference

INVITED TALKS

Efficient and Scalable Approach for Graph Machine Learning

- Invited Talk @ Korea-Japan Database (KJDB) Workshop, Nov. 2024

Knowledge Discovery from Real-world Networks: Dilemma, Methods, and Applications

- Invited Talk @ Data Intelligence Workshop, Jul. 2024
- Invited Talk @ New Faculty Session of KCC 2024, June. 2024
- Invited Talk @ Hanyang University, Feb. 2024

KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction

- Invited Talk @ Electronic & Information Research Information Center (EIRIC), Sep. 2023

Not All Layers Are Equal: A Layer-Wise Approach Towards Large-Scale DNN Training

- Poster Session @ Hyundai Vision Conference (HVC), Aug. 2023
- Invited Talk @ METU-HYU Joint Workshop, Dec. 2022

Basic Concept of Distributed Deep Learning with PyTorch Tutorials

- Invited Talk @ Medical AI Korea, Oct. 2021

Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach

- Invited Talk @ Waseda-UMS-Hanyang-UKM (WUHU) Joint Workshop, Dec. 2018
- Invited Talk @ NAVER Corp., Feb. 2018

TEACHING	Graduate School	
	CAU Data Mining (48768)	2025 Spring
	Underraduate School	
	- CAU Capstone Design (56120) - CAU Algorithm (13601) - CAU Database Design (34692) - CAU Artificial Intelligence (17182) - CAU Data Structure (40989)	Spring 2025 Spring 2024 - 2025 Fall 2024 Fall 2024 Spring 2024
PROFESSIONAL SERVICES	Track Co-Chair	
	ACM Symposium on Applied Computing (SAC)	2023 - 2025
	Program Committee Member (or Conference Reviewer)	
	- ACM International Conference on Information and Knowledge Management (CIKM) - ACM Web Conference (WWW) - ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) - IEEE International Conference on Data Mining (ICDM) - IEEE International Conference on Big Data (BigData), GTA3 Workshop - AAAI International Conference on Artificial Intelligence (AAAI) - ACM Symposium on Applied Computing (SAC)	2025 2023 - 2025 2021 - 2022, 2024 - 2025 2022, 2023 2023 2021 2022 - 2025
	Journal Reviewer	
	- ACM Transactions on Knowledge Discovery from Data (TKDD) - ACM Computing Surverys (CSUR) - IEEE Transactions on Neural Networks and Learning Systems (TNNLS) - IEEE Transactions on Network Science and Engineering (TNSE) - Journal of Supercomputing	2025 2024 2023, 2025 2024 2023
PATENTS	Granted Patents	
	- Asymmetric Centralized Training for Distributed Deep Learning Registration Number: KR10-2555268 - Multi-State Diffusion Model Using Interest, Intimacy, and Share Tendency Registration Number: KR10-2332348 - Accurate Ad-Effect Estimation Method based on Relevance between User and Item Registration Number: KR10-2144122 - Hybrid Approach for Influence Maximization in Social Networks Registration Number: KR10-1810864	Jul. 2023 Nov. 2021 Aug. 2020 Dec. 2017
	Filed Patents	
	- Personalized News Recommendation Technique using Category-guided Intent Disentanglement and Consistency-based News Representation Application Number: KR10-2023-0196006 - Political Stance Prediction Method and System using Political Knowledge Graphs and Hierarchical Attention Networks Application Number: KR10-2023-0059346 - Method and System for Adjusting the Learning Rate Differentiated by Layer for Large Scale Data Parallel Based Deep Learning Application Number: KR10-2022-0075800	Dec 2023 May 2023 June 2022